Assignment 4

(Q1-10) shuangli@link.cuhk.edu.cn, (Q11-20) yuqingzhou@link.cuhk.edu.cn

Please note that

- **Released date: 30th June, Fri.**
- **Due date: 7th July, Fri., by 11:59pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS1.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1 . Determine whether the following sets are bases for $\mathbb{R}^3$.

(a)
$$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$$

(b)
$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$$

(c)
$$\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$$

Question 2. Find a maximal linearly independent subset of the following set:

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} \right\}.$$

Question 3. Given

$$\mathbf{x}_1 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -9 \\ -2 \\ 5 \end{bmatrix}.$$

(a) Is $\mathbf{x} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$? why?

(b) Is $\mathbf{y} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$? why?

Question 4. Determine if the set $S = \{(x, y, z)^T | xy = z^2\}$ is a subspace of $\mathbb{R}^3$.

Question 5. Find the basis for

$$V = \text{Span}\left(\begin{bmatrix} 1 \\ 0 \\ -3 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 6 \\ -2 \\ -4 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}\right).$$

Question 6. Let

$$V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \middle| x_1 + x_2 + x_3 + x_4 = 0 \right\}.$$

By the definition of basis, show that

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

is a basis for V. Also find out the dimension of V.

Question 7. In each of the following, find the dimension of the subspace of P_3 spanned by the given vectors:

(a) $x, x-1, x^2+1$

(b) $x, x-1, x^2+1, x^2-1$

(c) $x^2, x^2-x-1, x+1$

(d) $2x, x-2$

Question 8. Let $B = \left\{ \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \end{bmatrix} \right\}$, $C = \left\{ \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\}$ be bases for $\mathbb{R}^2$.

Find the transition matrix from B to C and the transition matrix from C to B.

Question 9. Let $b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}, b_3 = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$.

(a) Show that the set $\mathbf{B} = \{b_1, b_2, b_3\}$ is a basis of $\mathbb{R}^3$.

(b) Given a vector $\mathbf{x} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}$ in $\mathbb{R}^3$. Find the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathbf{B}$.

(c) Given the coordinate of $\mathbf{y}$ with respect to the basis $\mathbf{B}$ is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. What is the vector $\mathbf{y}$ in $\mathbb{R}^3$.

Question 10. For each of the following choices of A and b, determine whether b is in the column space of A.

(a) $A = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$

Question 11. If $A = \begin{bmatrix} 1 & 2 & 3 & 5 & 0 & 2 & 4 \\ 2 & 1 & 3 & 4 & 7 & 8 & 9 \\ 1 & 1 & 2 & 3 & 2 & 2 & 5 \end{bmatrix}$

(a) Find a basis for $Col(A)$

(b) Find a basis for $Null(A)$ and find the dimension of $Null(A)$.

Question 12. Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ be linearly independent vectors in $\mathbb{R}^n$ and let

$$\mathbf{y}_1 = \mathbf{x}_2 - \mathbf{x}_1, \mathbf{y}_2 = \mathbf{x}_3 - \mathbf{x}_2, \mathbf{y}_3 = \mathbf{x}_3 - \mathbf{x}_1$$

Are $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ linearly independent? Prove your answer.

Question 13. Let

$$A = \begin{bmatrix} 1 & 2 & 2 & 3 & 1 & 4 \\ 2 & 4 & 5 & 5 & 4 & 9 \\ 3 & 6 & 7 & 8 & 5 & 9 \end{bmatrix}$$

Compute the reduced row echelon form U of A. Which column vectors of A correspond to the dependent variables of U? These column vectors form a basis for the column space of A. Write each of the remaining column vectors of A as a linear combination of these basis vectors.

Question 14. Let A be a 4×4 matrix with reduced row echelon form given by

$$U = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If

$$\mathbf{a}_1 = \begin{bmatrix} -3 \\ 5 \\ 2 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{a}_1 = \begin{bmatrix} 4 \\ -3 \\ 7 \\ -1 \end{bmatrix}$$

find $\mathbf{a}_3$ and $\mathbf{a}_4$.

Question 15. Let A be a 4×3 matrix and suppose that the vectors

$$\mathbf{z}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{z}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

form a basis for $N(A)$, If $\mathbf{b} = \mathbf{a}_1 + 2\mathbf{a}_2 + \mathbf{a}_3$, find all solutions of the system $A\mathbf{x} = \mathbf{b}$.

Question 16. Suppose $(x_1, x_2) + (y_1, y_2)$ is defined to be $(x_1 + y_2, x_2 + y_1)$. With the usual multiplication $c\boldsymbol{x} = (cx_1, cx_2)$, which of the eight axioms of a vector space (see Lecture 11) are **not** satisfied?

Question 17. Find a basis for the following solution set:

$$\left\{ \mathbf{x} \in \mathbb{R}^4 \middle| \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \mathbf{x} = 0 \right\}$$

Question 18. Let $\{x, 1\}$ and $\{2x - 1, 2x + 1\}$ be ordered bases for P_2, find the transition matrix representing the change in coordinates from $\{2x - 1, 2x + 1\}$ to $\{x, 1\}$

Question 19. Find a basis for following matrix:

$$\text{Null}\left(\begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}\right)$$

Question 20. For each of the following matrices, find a basis for the row space, a basis for the column space, and a basis for the null space.

(a)

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 4 & 7 & 8 \end{bmatrix}$$

(b)

$$\begin{bmatrix} -3 & 1 & 3 & 4 \\ 1 & 2 & -1 & -2 \\ -3 & 8 & 4 & 2 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$